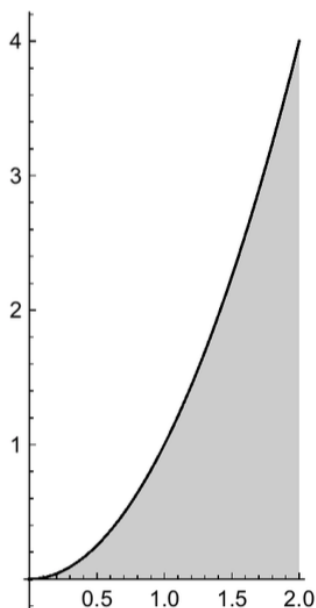


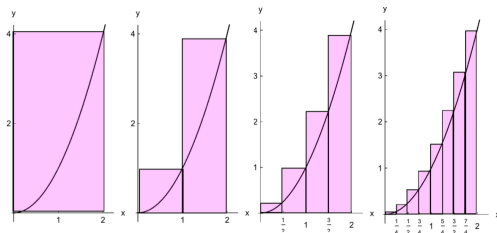
Problem 3

Problem 1 The graph of the function $f(x) = x^2$ is given in the figure.



Will a right Riemann sum approximation, for any value of n , be an underestimate or overestimate?

Approximate the shaded area using a right Riemann sum with $n = 1, 2, 4$, and 8 rectangles, as illustrated in the figure below.



- (a) Approximate the shaded area using a right Riemann sum with $n = 1$ rectangles.

Problem 3

- (b) *Approximate the shaded area using a right Riemann sum with $n = 2$ rectangles.*
- (c) *Approximate the shaded area using a right Riemann sum with $n = 4$ rectangles.*
- (d) *Approximate the shaded area using a right Riemann sum with $n = 8$ rectangles.*